

Healthcare Appointment No-Show Prediction

Internship Project Report

Abstract

Missed healthcare appointments represent a widespread operational challenge, resulting in wasted clinical resources, longer patient wait times, and increased costs. This project leverages data visualization to uncover behavioral and demographic patterns contributing to appointment no-shows. Using a real-world dataset from Brazil, the goal was to explore correlations between no-shows and factors such as patient age, gender, weekday of the appointment, and SMS reminder notifications. The final outcome is a visually rich and interactive dashboard developed in Power BI that enables decision-makers to interpret trends and optimize patient scheduling strategies.

Introduction

In healthcare systems worldwide, patient no-shows are a persistent problem that compromises care continuity and clinic efficiency. Understanding and predicting which patients are most likely to miss their appointments can allow healthcare providers to proactively intervene—through reminders, rescheduling strategies, or follow-up communications. This project seeks to identify such patterns using visual analytics. The dataset used contains over 100,000 records of medical appointments, including whether or not a patient attended. By exploring these trends through a creative Power BI dashboard, the project offers data-backed recommendations to reduce absenteeism and improve healthcare delivery outcomes.

Tools Used

Tool	Purpose
Power BI	Building dashboards and visualizations
Python (Jupyter Notebook)	Initial data exploration and transformation
Pandas & Matplotlib	Quick validation of relationships
Microsoft Excel	Previewing and organizing column fields

Steps Involved in Building the Project

1. Data Import & Schema Review

The dataset, `KaggleV2-May-2016.csv`, was loaded directly into Power BI. It includes columns like Gender, Age, ScheduledDay, AppointmentDay, No-show, and SMS_received.

2. Feature Engineering & Field Creation

- A new AgeGroup column was created using custom ranges (e.g., 0–18, 19–35, etc.).

- A Weekday column was derived from AppointmentDay using DAX formulas to identify trends by day.
- The No-show column was recoded for filtering ("Yes" = missed).

3. Dashboard Visuals

- **KPI Cards:** Displayed total appointments (22,319), total no-shows, and calculated no-show rate.
- **Bar Chart:** Showed no-show counts by age group.
- **Donut Chart:** Compared no-shows by gender (higher in females).
- **Line Chart:** Visualized weekday trends (highest on Tuesdays).
- **Matrix Table:** Combined age and gender breakdown.
- **Stacked Bar Chart:** Highlighted the impact of SMS reminders.

4. Aesthetic Design

- Gradient purple background for visual appeal
- Uniform color coding and font styling
- Clean, centered title: **"Healthcare Appointment No-Show Prediction"**
- Structured layout separating KPIs, demographics, and behavior trends

Key Insights

- **Young Adults (19–35)** and **Teens (0–18)** accounted for the highest number of missed appointments.
- **Females** were more likely to miss appointments than males.
- **Tuesday** saw the highest no-show rates, followed by Wednesday and Monday.
- Patients who **did not receive SMS reminders** were more likely to skip their appointments.
- The **overall no-show rate** is significant, emphasizing the need for scheduling improvements.

Conclusion

This project successfully delivers a fully interactive Power BI dashboard that visualizes patient behavior patterns affecting appointment attendance. Through a combination of visual storytelling and analytical breakdowns, it offers insights that can be translated into actionable scheduling policies. Healthcare institutions can use such tools to identify high-risk groups, improve communication strategies (e.g., SMS reminders), and better allocate appointment slots—ultimately improving patient turnout and operational efficiency. The project demonstrates the power of data visualization in addressing real-world healthcare challenges.